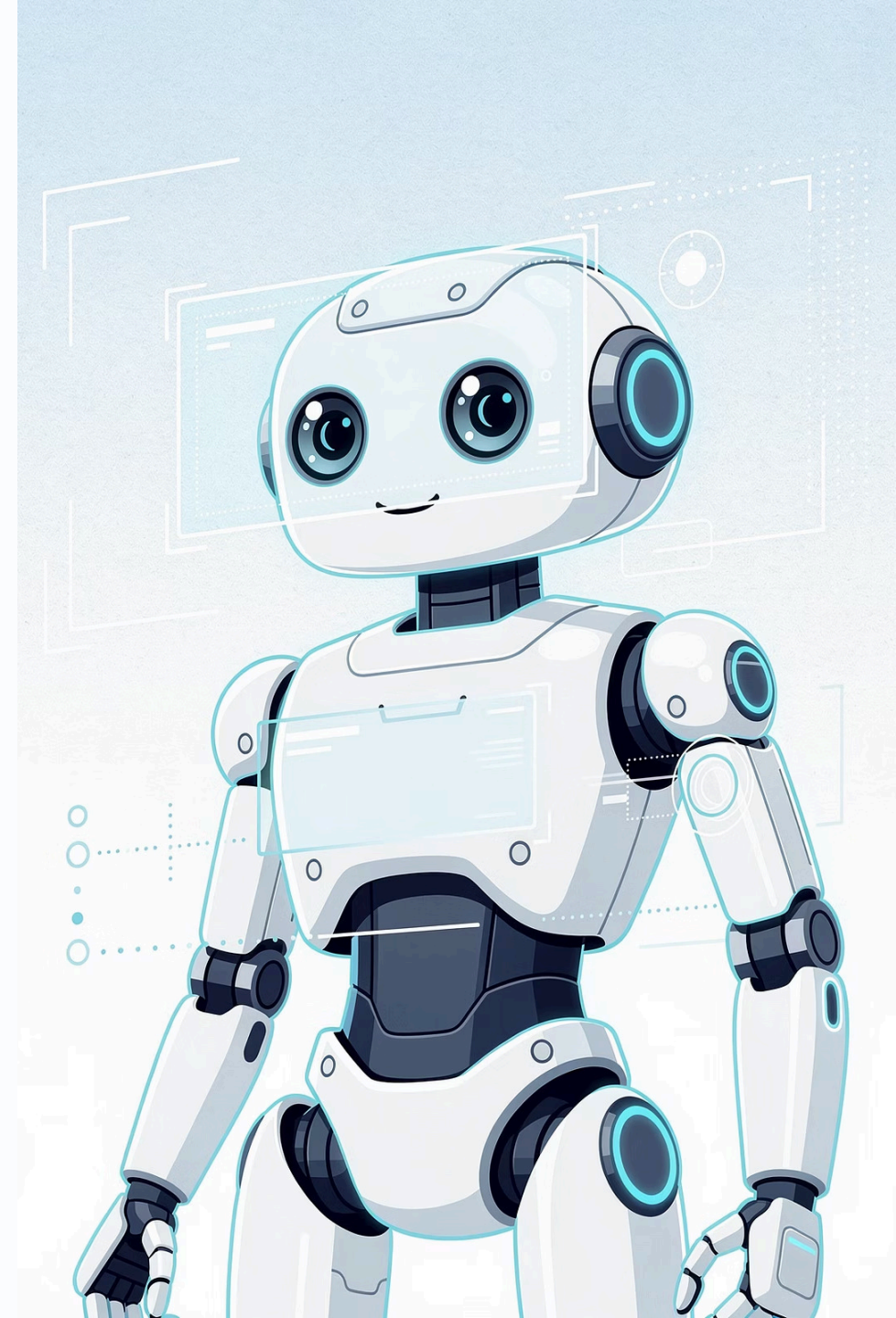


Session 15:

AI That Sees - Computer Vision

Last Session, you saw how AI **reads**. Now we'll find out how AI **SEES**.

- How does your phone unlock with your face?
- How does a self-driving car know there's a stop sign?
- How does Snapchat put a flower crown on your head perfectly?



Objectives

By the end of this session, you'll be able to:

01

Define Computer Vision

Explain what Computer Vision is and where it shows up in your life.

02

Know the 3-Step Recipe

See → Process → Act - the core loop behind all CV systems.

03

Understand Pixels

Know what pixels are - the building blocks of every image AI sees.

04

Spot CV Jobs

Identify face recognition, object detection, OCR, and more in the wild.

05

Know What Trips AI Up

Understand why lighting, angles, and hidden things confuse AI's eyes.



Recap & Homework

Welcome back! Let's check in.

Share your **NLP Detective** examples! Where did you spot NLP this week in the real world?

Quick Check from Last Week

- What's NLP?
- What's the 3-step recipe?
- Which step does AI struggle with the most?

Warm-Up - Seeing Is Tricky Too!

Look at these everyday situations. You handle them easily - but for a computer? Way harder.



Hat & Sunglasses

A photo of your friend wearing a hat and sunglasses.

Is that still your friend?



Snowy Stop Sign

A stop sign covered in snow.

Is it still a stop sign?



Cat on Carpet

Your cat sitting on a really patterned carpet.

Where does the cat end and the carpet begin?

 You see these things easily. But for a computer? **Way harder.** That's exactly what we're exploring today.

What is Computer Vision?

Computer Vision (CV) = AI that works with PICTURES and VIDEOS.

You use CV all the time:

Face unlock

Google Photos
sorting

Snapchat filters

Most Computer Vision follows 3 simple steps:



SEE

Take in pixels (the image)



PROCESS

Figure out what's in the picture



ACT

Do something with what was seen

It's like the NLP recipe - just for **eyes** instead of words!



Step 1 - SEE

To AI, an image isn't really an image. It's a **HUGE grid of tiny dots** called **PIXELS**.

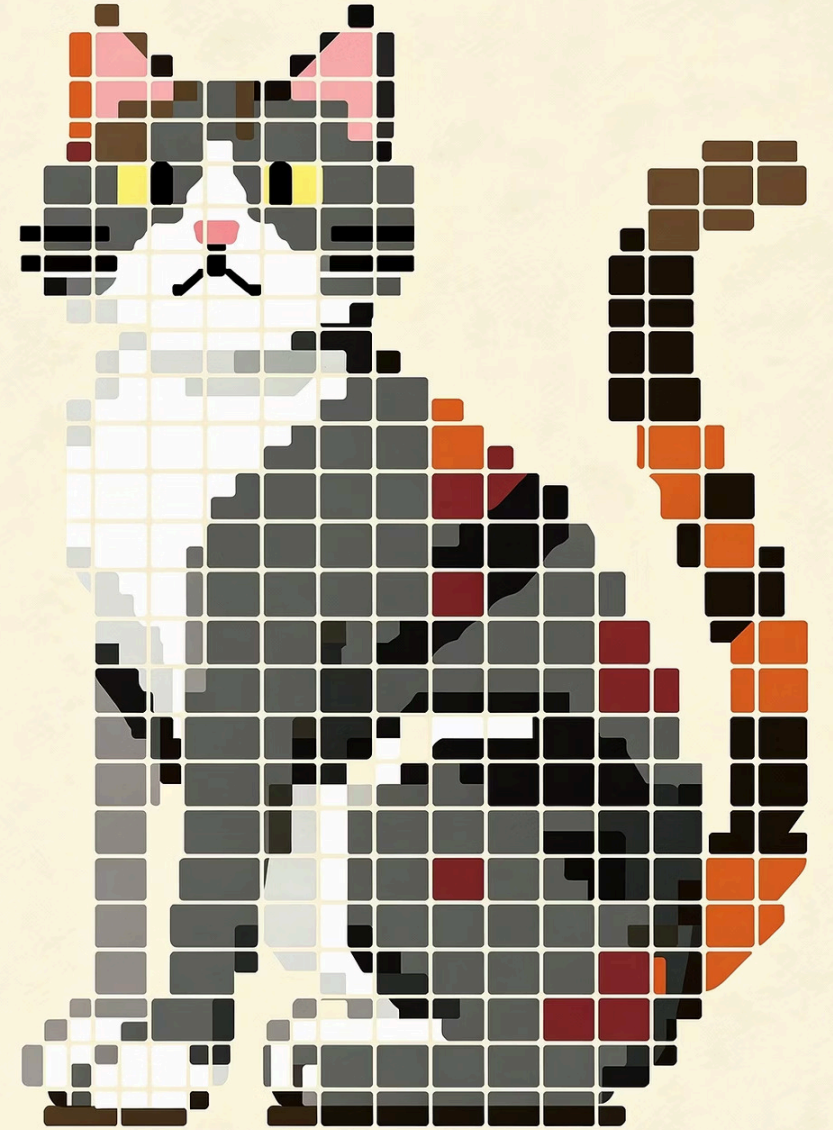
What is a Pixel?

Every photo on your phone is just **millions of pixels**, each one a tiny colored square arranged in a grid.

A normal phone photo? About **12 million pixels**!

If you counted them out loud - one per second - it would take you **almost 5 months without sleep**.

- ❏ AI doesn't see "a cat." It sees: pixel #1 is brown, pixel #2 is brown, pixel #3 is brown... and so on, **12 million times**.



Each Pixel Is Just 3 Numbers

Every pixel = **3 numbers: Red, Green, Blue** (RGB). Mix the 3 numbers → you get any color in the world.

That's all an image is to AI - just numbers. Millions and millions of numbers.

 **Red**
How much red? (0 to 255)

 **Green**
How much green? (0 to 255)

 **Blue**
How much blue? (0 to 255)

Color	RGB Values	Result
Pure Red	(255, 0, 0)	
Pure Green	(0, 255, 0)	
Pure White	(255, 255, 255)	
Pure Black	(0, 0, 0)	



Mini Activity 1 - Be a Pixel Detective!

Let's see pixels in action. Open any photo on your phone or computer.

Now **ZOOM IN** - really zoom in. What happens?

01

Open a Photo

Pick any photo - a selfie, a pet, anything.

02

Keep Zooming In on a Face

Zoom in as far as your device allows.

03

Watch It Change

At some point - it stops looking like a face.

04

See the Squares!

You see colored squares. Those are pixels!



That's exactly what AI sees ALL the time. It never sees a face - only the squares.

Step 2 - PROCESS 🧠

Now AI has the pixels. What does it do with them? It looks for **patterns** - step by step, building up what it sees from millions of tiny clues. Like solving a puzzle, one tiny piece at a time.

Edges

Where colors suddenly change - this creates an outline of objects.

Shapes

Circles, lines, curves - the building blocks of recognizable objects.

Colors

What colors are next to each other - context clues for identification.

Familiar Things

"This pattern looks like an eye!" - matching to learned patterns.

How AI "Sees" a Face

Imagine AI looking at a photo of your face. It doesn't see YOUR face like you do - it sees a **pattern of pixels** that matches what it's learned a face looks like. Same way you'd recognize an old friend just from their silhouette.

Find Edges

Outline of your head

Eye Shapes

Two dark ovals detected

Nose

Shape in the middle

Mouth

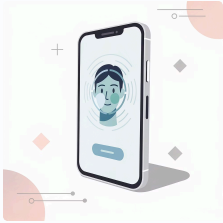
Shape below the nose

"FACE!"

Pattern matches = face recognized

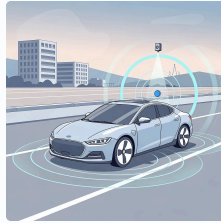
Step 3 - ACT

Once AI has processed what it sees - it **does something**. The action is the whole point of the system.



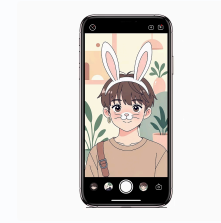
Face Unlock

"That's you! Phone unlocks."



Self-Driving Car

"Stop sign ahead! Apply brakes."



Snapchat Filters

"Found a face! Add bunny ears here."



Google Photos

"That's a cat! Put it in the 'cats' folder."



Cancer Detection

"Suspicious spot. Alert the doctor."

Common CV Jobs

Computer Vision does TONS of jobs.

CV Job	What it does	Example
 Image Classification	"What IS this?"	Cat or dog?
 Object Detection	"Where ARE things?"	Box around each car in a photo
 Face Recognition	"WHO is this?"	Unlock phone with your face
 OCR	Reads text in images	Snapping a sign and reading it
 Scene Understanding	"What's HAPPENING?"	Self-driving car: red light + person crossing
 Anomaly Detection	Finds what shouldn't be there	Tumor on X-ray, bruise on a strawberry

Mini Activity 2 - Match the CV Job!

Look at each example. What CV job is happening? Try to match before you see the answers!

What's happening	CV Job?
Your phone unlocks when you look at it	?
Google Translate camera reads a sign in Spanish	?
A car detects a pedestrian and stops	?
Instagram suggests "tag your friend" on photos	?
Google Photos sorts your trip into "Beach Day"	?
Snapchat puts a flower crown on your head	?

 **Options:** Face Recognition / Object Detection / OCR / Image Classification / Scene Understanding

Mini Activity 2 - Answers

What's happening	CV Job
Phone unlocks with your face	 Face Recognition
Translate camera reads sign	 OCR
Car detects pedestrian	 Object Detection
Tag your friend suggestion	 Face Recognition
Google Photos sorts your trip	 Image Classification
Snapchat flower crown	 Face Recognition <i>(it has to find your face first!)</i>

- ☑ See how many different CV jobs are working at once? Your phone unlocking uses **MULTIPLE CV jobs** - that's how it's so reliable.

Why Is Seeing HARD for AI?

Even with the 3-step recipe, AI gets confused. That's why the **Process step is the hardest** - there's SO MUCH to figure out.

Lighting

Same person, dark room vs sunny day → looks totally different to AI.

Angles

Your face from front vs from side → AI might not recognize it's still you.

Occlusion

Wearing a mask or sunglasses → AI loses important clues.

Bias in Data

If AI was trained mostly on light-skinned faces, it works worse for darker-skinned faces.

Motion Blur

Fast-moving objects → AI can't focus on the pixels clearly.

Cluttered Backgrounds

Busy patterns confuse the Process step - like a cat on a patterned carpet!

Face Recognition Bias

True Story from 2018

Researchers tested major face recognition tools on hundreds of faces. They found a **huge gap** in accuracy:

✓ Light-Skinned Men

Very few mistakes - high accuracy across the board.

✗ Dark-Skinned Women

Mistakes up to **1 in 3 times!** - dramatically worse performance.

Why? Because the training data was mostly light-skinned faces. The AI never learned to "see" everyone equally. This is **bias** - and it matters because face recognition is being used in real places: phones, security systems, photo apps, and more.

⚠ **Better data + more diversity = more fair AI. Always.**



Hands-On Activity - Computer Vision in Action

Time to play with real Computer Vision tools! We're going back to a tool you've already used - **Google Teachable Machine**. But this time, we're going to think about it **differently**.

Train a Tiny Model

But pay close attention to **WHAT it's actually seeing** - not just whether it works.

Try to Break It on Purpose

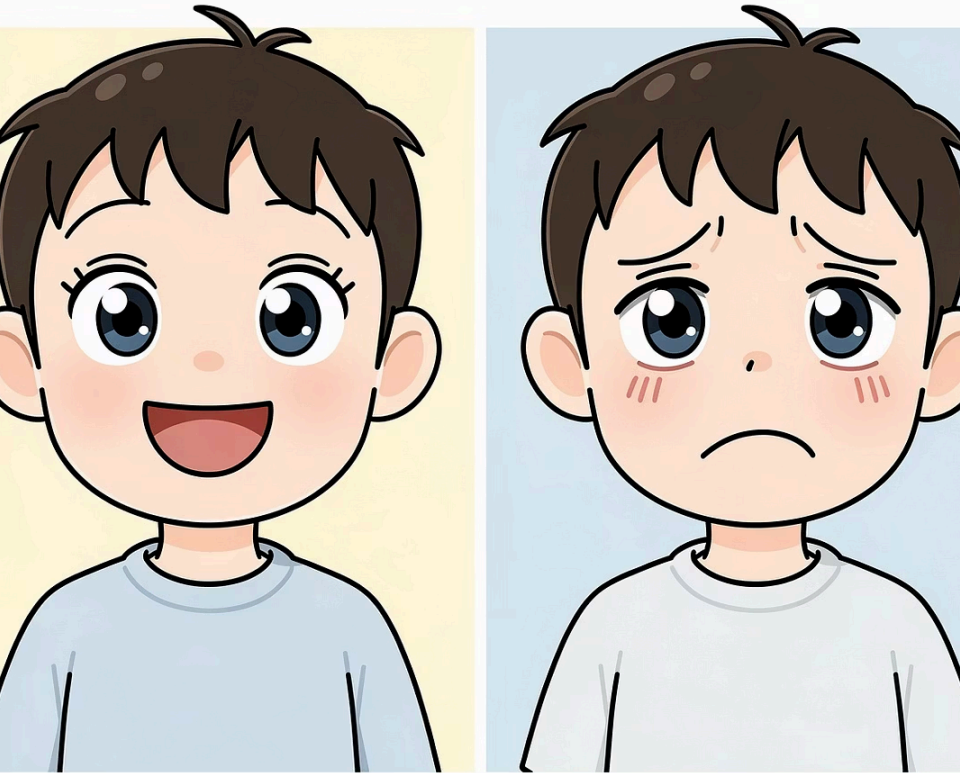
Test the edges of what the model can handle.

Spot Where Process Fails

Notice exactly which step breaks down and why.

Mini Build - Happy vs Sad Face! 😊

Train Teachable Machine to recognize happy and sad faces. Here's how:



Capture Happy Faces

Take 15–20 photos of yourself making a **HAPPY face** (smile) - try different angles and lighting.

Capture Sad Faces

Take 15–20 photos making a **SAD face** (frown) - same variety of angles.

Train the Model

Hit train and watch the AI learn from your pixel data.

Test It!

Show your face and see what it predicts - with a confidence score.

① **SEE** = camera captures pixels.

PROCESS = AI looks at edges, mouth shape, eye shape

ACT = Shows "Happy" or "Sad" with confidence score

Try to BREAK Your Model! 🌟

Now the fun part, see where Computer Vision fails! Try these and watch what happens. Take notes on what broke it!

🕶️ Sunglasses or Hat

Does it still work? (*Occlusion!*)

💡 Dark Corner

Move to a dim spot. Does it still work?

📐 Tilt Your Head

Sideways or upside down. Does it still work?

😡 Angry Face

You didn't train this! What does it predict? (*Unfamiliar data!*)

🙌 Cover Half Your Face

With your hand. Does it still work? (*Partial occlusion!*)

What Did You Notice? 🤔

Discussion Questions

- What broke your model the easiest?
- Was the **Process** step really doing what you thought?
- If you wanted to make it BETTER - what would you add to your training data?

Bigger Questions

- Is your tiny model fair to ALL faces?
- What if you trained it on only one expression?
- Why does big AI need MILLIONS of photos to be reliable?

📌 Your tiny model is a window into how ALL computer vision works - the same challenges exist at every scale.

What We Learned in this Session

Computer Vision (CV)

AI that works with images and videos - it's everywhere in your daily life.

3-Step Recipe

See → **Process** → **Act** - the core loop behind every CV system.

Pixels & RGB

Pixels are tiny colored dots - the LEGO bricks of images. Every pixel = 3 numbers (red, green, blue).

CV Jobs

Classification, Object Detection, Face Recognition, OCR, Scene Understanding, Anomaly Detection.

What Trips Up CV

Lighting, angles, occlusion, bias, blur, and cluttered backgrounds.

Built Your Own CV Model

Trained and tested a real model in Teachable Machine - and broke it on purpose!

Your Homework - CV Detective

Spot Computer Vision in your daily life this week. **Mission:** Find **3 examples** of CV being used somewhere around you.



Face Unlock / Face ID

Every time you unlock your phone with your face.



Snapchat Filters

Any AR filter that tracks your face or surroundings.



QR Code Reader

Scanning a QR code is OCR in action!



Self-Checkout Scanning

The store scanner reads barcodes and product images.

What	CV Job	Tool/App
Example 1		
Example 2		
Example 3		

Next Time: Session 16

AI That Hears & Speaks - Voice AI

You've learned how AI reads. You've learned how AI sees. Next session: how does AI **HEAR** and **TALK**?



Siri Understands You

How does AI turn your voice into meaning?



AI Speaks Back

How can AI speak in a voice that sounds human?



AI-Generated Voices

What's an AI-generated voice? We'll make one!



Amazing & Risky

Why are AI voices both incredible and dangerous?

